Network Science

Lecture 09: Node Representations and Graph Embeddings

Prof. Dr. Michael Granitzer

Dr. Kanishka Ghosh Dastidar

Dr. Jelena Mitrovic

Speaker notes

This lecture closes the arc that opened with L07. L07 asked: "given a graph, can we navigate it with local information?" Today we ask the dual question: "given a graph, can we turn each node into a *vector* that captures its role, neighbourhood, and community — so that classical ML, retrieval, and reasoning operate directly on nodes?" The two lectures together form the embedding-and-navigation pipeline that powers modern graph ML, recommender systems, and retrieval-augmented LLMs. The lecture has a clear narrative: spectral methods (principled but global), random-walk embeddings (scalable, unsupervised), GNNs (supervised, feature-aware) — each overcomes a limitation of the previous step.

From Structure to Representation

Every previous lecture analysed the graph as a **combinatorial object**: edges, triangles, communities, cuts, shortest paths. Today we ask the dual question:

> **Can every node be represented as a *point* in $\mathbb{R}^d$ such that graph structure is preserved as geometry?**
>
> If two nodes are "similar" in the graph (same community, similar role, co-visited by walks), they should be *close* as vectors.

Speaker notes

The conceptual shift is the same one that word2vec introduced for language in 2013: instead of manipulating symbols, manipulate vectors that carry the structural signal. For networks, "structural signal" can mean many things — community membership, role, shortest-path proximity, neighbourhood composition — and the different embedding methods differ primarily in *which* of these they preserve.

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This is **graph representation learning** — turning a discrete object into a continuous one so that we can use every tool from classical ML: clustering, classification, retrieval, generative models.

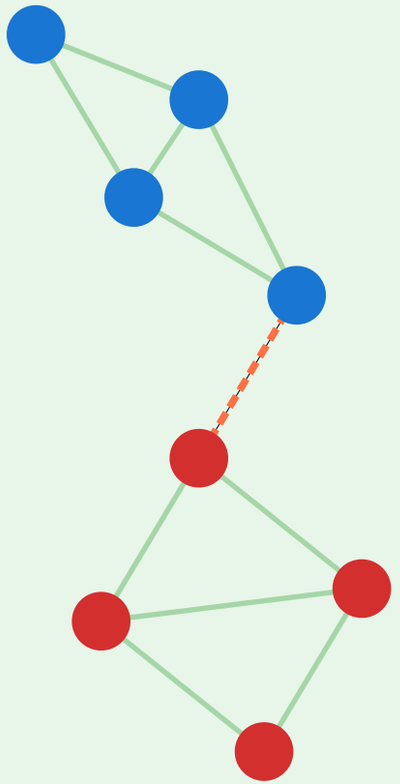
The Question — Made Visual

Laplacian Eigenmaps — From Graph to Geometry

Eigenvectors of the Laplacian embed each node as a point in $\mathbb{R}^d$

1. Graph G

two weakly connected clusters



colour = sign of Fiedler vector

2. Laplacian L

$L = D - A$

$$\begin{bmatrix} 2 & -1 & 0 & 0 & \cdots \\ -1 & 3 & -1 & -1 & \cdots \\ 0 & -1 & 3 & -1 & \cdots \\ 0 & -1 & -1 & 2 & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

eigendecomposition

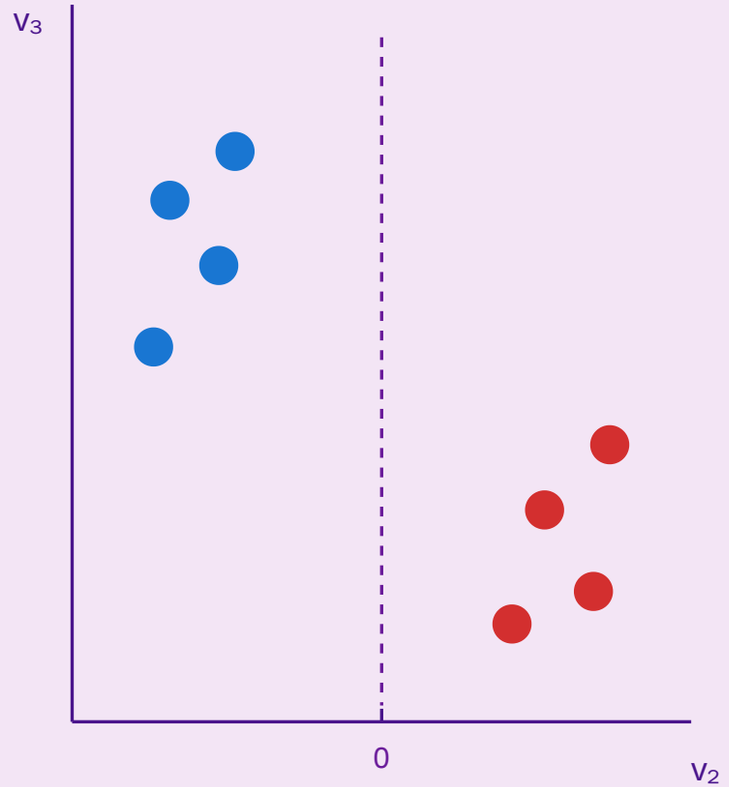
$L v_k = \lambda_k v_k$

$0 = \lambda_1 \le \lambda_2 \le \lambda_3 \le \dots$

take v_2, v_3 as coords

3. Embedding $\mathbb{R}^2$

$z_v = (v_2(v), v_3(v))$



k-means on z gives clusters
= spectral clustering

A graph with two weakly connected clusters (left) is mapped through the Laplacian (middle) to $\mathbb{R}^2$ (right) such that the two clusters separate linearly. The coordinates are the second and third eigenvectors of the Laplacian — a *continuous* summary of a *discrete* partition.

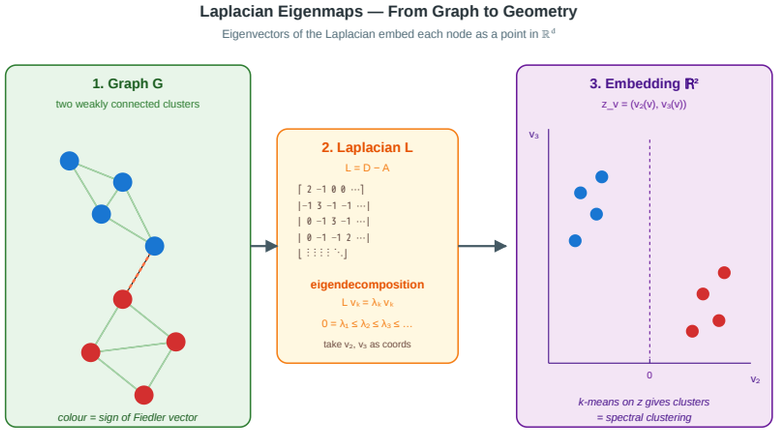
Speaker notes

This figure is the lecture anchor. It shows in three panels what we are trying to achieve: the graph has visible structure (two clusters joined by a bridge). The Laplacian encodes this structure in matrix form. Its low eigenvectors place each node in $\mathbb{R}^2$ so the clusters separate. Every method in today's lecture — spectral, DeepWalk, node2vec, GNNs — is an attempt to compute this kind of map, differing in what "structure" they preserve and how they scale.

A Sixth Gap — The Representation Gap

L07 (navigability) and L09 (representation) are the two faces of the same problem: making graphs *work* with local, vector-based operations.

Lecture	Gap	Strategy
L07	Navigational	Multi-scale links
L09	**Representation**	**Embed nodes in $\mathbb{R}^d$**
L08	Process-structure	Compare processes



Speaker notes

The representation gap is the ability of a method to express graph structure in a form that downstream classifiers, clusterers, retrievers, or LLMs can consume. L07 made structure *traversable*; L09 makes structure *computable*. Together they form the two pillars of modern graph-ML infrastructure: HNSW indexes (L07) built on top of learned node embeddings (L09).

Takeaway

Graph representation learning turns a discrete object into a continuous one. The goal is a map $z : V \to \mathbb{R}^d$ such that graph similarity $\approx$ vector similarity. Different methods capture different notions of "similar."

Spectral Foundations

Guiding Question: Can the eigenvectors of a graph matrix give us a natural geometry for its nodes?

Speaker notes

This module re-introduces the graph Laplacian (seen in L04 for community detection) and reframes it as the first — and still foundational — node-embedding technique. Spectral methods give a principled, globally optimal answer to the embedding question, but they scale poorly and have no notion of node features. That tension motivates the rest of the lecture.

The Graph Laplacian — One More Time

For an undirected graph $G = (V, E)$ with adjacency A and degree diagonal D:

$$L = D - A \qquad \text{(combinatorial Laplacian)}$$

$$L_{\text{sym}} = I - D^{-1/2} A D^{-1/2} \qquad \text{(normalised Laplacian)}$$

L is symmetric positive semi-definite. Its eigenvalues satisfy $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$.

Speaker notes

The Laplacian was introduced in L04 for spectral clustering via the Fiedler vector. Today we generalise that usage: instead of one eigenvector for one cut, we stack the first d non-trivial eigenvectors into a matrix, and each node's row is its embedding. This is *Laplacian Eigenmaps* (Belkin & Niyogi 2003), one of the first principled node-embedding methods.

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The eigenvectors corresponding to the smallest non-zero eigenvalues are the most informative: they encode the graph's *large-scale* structure.

Laplacian Eigenmaps (Belkin & Niyogi 2003)

Let $v_2, v_3, \ldots, v_{d+1}$ be the eigenvectors of L for the d smallest non-zero eigenvalues. Define

$$z_v = \big(v_2(v), v_3(v), \ldots, v_{d+1}(v)\big) \in \mathbb{R}^d$$

Each node gets a d-dimensional embedding.

Why this works. The embedding is the solution to

$$\min_{Z} \sum_{(u,v)\in E} \|z_u - z_v\|^2 \quad \text{s.t. } Z^\top D Z = I$$

— connected nodes end up close, and the constraint prevents the trivial solution $z = 0$.

Speaker notes

The variational interpretation is important: Laplacian Eigenmaps is the *exact* solution to the most natural embedding objective — minimise squared distance between neighbours, under an orthogonality constraint. The first eigenvector v_1 is constant and uninformative (it corresponds to $\lambda = 0$); we start at v_2, the Fiedler vector. So d-dimensional embeddings use eigenvectors $v_2 \dots v_{d+1}$.

Spectral Clustering = k-means on Laplacian Eigenmaps

Laplacian eigenmaps + k-means = **spectral clustering**.

- Nodes in the same community have similar low-frequency eigenvector values.
- k-means on $Z \in \mathbb{R}^{|V| \times d}$ recovers communities.
- This is how NCut (Shi & Malik 2000) and most image-segmentation algorithms work.

Spectral gap. The jump $\lambda_{k+1} - \lambda_k$ indicates the "natural" number of communities. A large gap means the graph has *exactly* k clear clusters; a small gap means the partition is ambiguous.

Speaker notes

This slide connects L04 (community detection) to L09 (representation). The Fiedler vector we used in L04 for the binary cut is the same v_2 we use today as the first coordinate of the embedding. Going from one eigenvector to d eigenvectors generalises from binary to k-way clustering, and from clustering to clustering to *embedding* (a richer object that can feed any downstream task, not just clustering).

Signed Laplacian — Bridge to L06

For **signed networks** (L06, structural balance), the *signed Laplacian* $\bar{L} = \bar{D} - A$ (with $\bar{d}_i = \sum_j |A_{ij}|$) encodes enmity as well as friendship.

Low eigenvectors of $\bar{L}$ embed nodes such that friends cluster and enemies separate — the continuous analogue of approximate structural balance.

Speaker notes

This is a hidden link between L06 and L09. Structural balance gave us discrete, combinatorial tests for when a signed graph can be two-clustered. The signed Laplacian gives us a continuous relaxation: every graph has a signed-spectral embedding, and how *balanced* the graph is determines how cleanly the embedding separates. For students who remember L06's frustration with NP-hardness in imperfect balance: here is the relaxation that makes the problem tractable.

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Kunegis et al. (2010) used this to detect communities in signed online networks (Slashdot, Epinions) where positive and negative edges coexist.

The Spectral Limitations

Why can't we just use spectral embeddings for everything?

1. **Cost.** Computing eigenvectors of a $10^9 \times 10^9$ Laplacian is infeasible. Best sparse solvers still take $O(|E| \cdot d)$ per iteration.
2. **Transductive.** A new node requires recomputing the whole spectrum — there is no "inference" step.
3. **No features.** Spectral embeddings use only graph structure; they ignore node attributes (text, images, tabular).
4. **Global.** All nodes see the same basis; local neighbourhood-specific patterns get averaged away.

Speaker notes

These four limitations are why modern graph-ML moved beyond pure spectral methods. Each subsequent module in this lecture addresses one of them: random-walk methods (2) and (3) partially, GNNs fix (2)-(4) entirely. But spectral embeddings remain the *gold standard* for small-to-medium graphs and the *theoretical benchmark* against which newer methods are compared.

Takeaway

> Spectral methods embed each node using the graph Laplacian's low eigenvectors. They are principled, globally optimal, and reduce clustering to linear algebra — but they scale poorly, are transductive, and ignore node features.

Quick Check

You have a graph with 3 disconnected components of similar size.

1. What are the first three eigenvalues of the Laplacian L?
2. What do the first three eigenvectors tell you?
3. What does "spectral gap" mean for this graph?

Speaker notes

Give them 3 minutes. (1) $\lambda_1 = \lambda_2 = \lambda_3 = 0$. (2) They span the space of constant-per-component vectors — each indicator of a component is a linear combination of these three eigenvectors. (3) The spectral gap here is between $\lambda_3 = 0$ and $\lambda_4 > 0$. A large λ_4 means the three components are well-separated *and* internally well-connected; a small λ_4 means at least one component is itself close to disconnecting.

Quick Check — Solution

$\lambda_1 = \lambda_2 = \lambda_3 = 0$ — the multiplicity of $\lambda = 0$ equals the number of connected components.

They span the space of piecewise-constant vectors (constant on each component). Any indicator vector of one component is a linear combination of them.

The gap is between $\lambda_3 = 0$ and λ_4. Large $\lambda_4 \rightarrow$ components are well-separated and internally well-knit. Small $\lambda_4 \rightarrow$ one component is close to splitting further.

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Random Walks as Representations — DeepWalk

Guiding Question: Can we use *samples* of graph structure (random walks) instead of spectral factorisation?

Speaker notes

DeepWalk (Perozzi et al. 2014) is the first scalable node-embedding method. Its innovation is treating a random walk on the graph as a "sentence" whose "words" are nodes, then applying word2vec's skip-gram model unchanged. The result is an unsupervised method that scales to millions of nodes and trains with stochastic gradient descent instead of eigendecomposition.

The Analogy — Graph ≈ Corpus

Language (word2vec)	Graph (DeepWalk)
Vocabulary	Node set V
Corpus	Collection of random walks
Sentence	One walk $v_1 v_2 \dots v_L$
Word	Node v_i
Context window	Walk window of size c
Skip-gram objective	Skip-gram objective

Speaker notes

This is the key conceptual move. In 2013, Mikolov et al. showed that skip-gram with negative sampling trains n-dimensional word vectors by predicting neighbouring words. Perozzi, Al-Rfou & Skiena saw that if they interpreted a random walk as a sentence, every property that made word2vec work — locality, co-occurrence as signal, scalable SGD training — transfers directly to graphs. No new optimisation machinery is needed.

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Context window	Walk window of size c
Skip-gram objective	Skip-gram objective

Insight. If we can generate many random walks from each node, we can reuse the entire word2vec machinery to learn node embeddings.

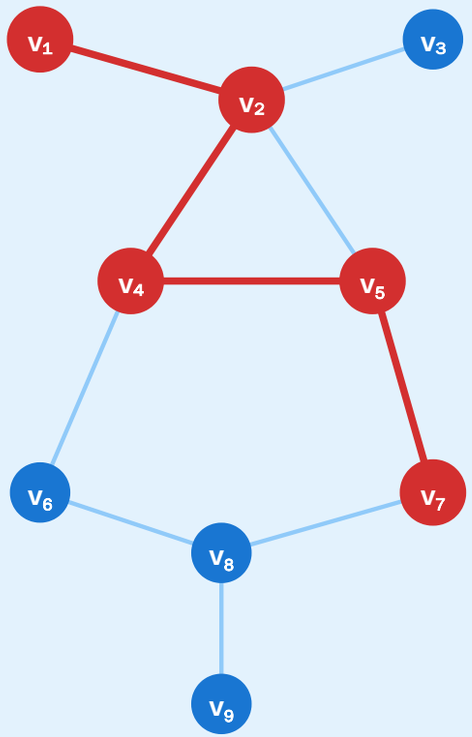
DeepWalk Pipeline

DeepWalk — Random Walks as "Sentences" for Skip-Gram

node ≈ word · walk ≈ sentence · co-occurrence → embedding

1. Random walks

length L from each node



walk: $v_1\ v_2\ v_4\ v_5\ v_7$

repeat y times per node

2. Skip-gram window

context c = 2 around centre



centre v_4 · context $\{v_1, v_2, v_5, v_7\}$

training pairs (v, context)

(v_4, v_1) (v_4, v_2)

(v_4, v_5) (v_4, v_7)

Skip-gram objective

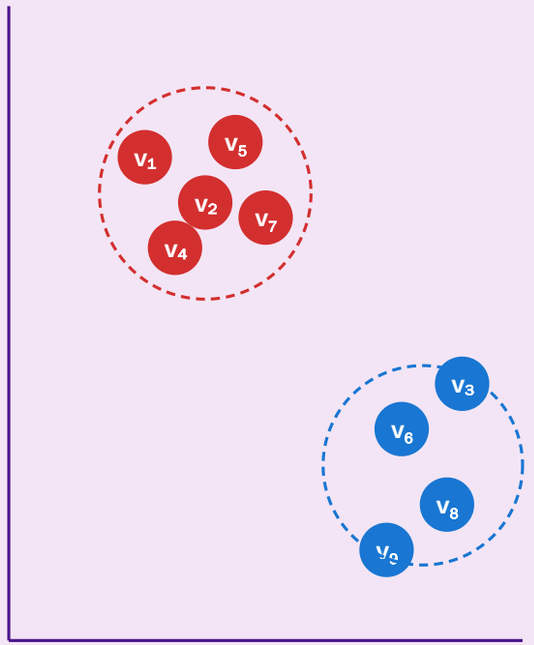
$$\max \Sigma \log P(u \mid v)$$

$$P(u|v) = \text{softmax}(z_u \cdot z_v)$$

"co-walking nodes have similar z"

3. Embeddings

$z_v \in \mathbb{R}^d$, $d \ll |V|$



co-walked nodes end up close in the embedding space

Perozzi et al. 2014

Speaker notes

The figure shows the entire pipeline in three panels — graph, skip-gram window, embedding. This is the textbook diagram for random-walk embeddings.

DeepWalk uses *uniform* random walks: at each step, choose a uniformly random neighbour. In the next module we will see that this uniform choice is a limitation — biasing the walk with two parameters turns DeepWalk into node2vec.

(1) From each node, sample γ random walks of length L. (2) Slide a window of size c over each walk; emit $(v, \text{context})$ training pairs. (3) Train skip-gram to maximise $\log P(u \mid v)$ for co-walked pairs. Nodes that appear together in walks end up close in the embedding space.

The Skip-Gram Objective

Given walks $\{W_i\}$ and window c, learn vectors $z_v \in \mathbb{R}^d$ maximising

$$\mathcal{L} = \sum_i \sum_t \sum_{-c \le j \le c, j \neq 0} \log P(v_{t+j} \mid v_t)$$

with

$$P(u \mid v) = \frac{\exp(z_u \cdot z_v)}{\sum_{u' \in V} \exp(z_{u'} \cdot z_v)}$$

Dense softmax over $|V|$ is too expensive → **negative sampling** approximates it by k random non-context nodes per pair.

Speaker notes

Negative sampling turns the full softmax into a binary classification: distinguish true co-walk pairs from random "negative" pairs. Typical $k = 5$–20. Training is SGD with embeddings z_v and a second set of "output" vectors for context nodes. After training, z_v is the node embedding. This is exactly what word2vec does; the only difference is where the training pairs come from.

What DeepWalk Captures

Proposition (Qiu et al. 2018). DeepWalk with long walks is *implicitly factorising* the log of a shifted PPMI matrix built from the random-walk co-occurrence matrix.
This connects DeepWalk back to spectral methods: both minimise a matrix-factorisation objective — the matrices just differ.

Key result. Random-walk methods and spectral methods sit on the same theoretical spectrum. DeepWalk is an approximate, sample-based spectral embedding of the walk co-occurrence matrix.

Speaker notes

This is one of the most beautiful theoretical results in graph-ML. Qiu et al. (WSDM 2018) proved that DeepWalk, node2vec, LINE, and PTE all reduce to matrix factorisation of different *co-occurrence matrices* built from the graph. This unifies the "spectral" and "random-walk" families: they differ only in which matrix they factor and how they do it (exact eigendecomposition vs. SGD on samples). Practical implication: DeepWalk scales because SGD on samples is cheaper than eigendecomposition — but the mathematical object being computed is the same kind of matrix factorisation.

Takeaway

DeepWalk treats a random walk as a sentence and reuses word2vec's skip-gram. It is scalable (SGD, no eigendecomposition) and unsupervised. Under the hood, it is factorising a co-occurrence matrix — the same machinery as spectral embedding, in a sampled form.

node2vec — Biased Walks

Guiding Question: Should "similar" mean *same community* or *same structural role*? Can one embedding capture both?

Speaker notes

node2vec (Grover & Leskovec 2016) asks a sharper question than DeepWalk: what kind of graph similarity should the embedding capture? It answers by introducing *biased* random walks with two parameters, p and q, that interpolate between BFS-like (structural) and DFS-like (community) exploration.

Two Notions of Similarity

Homophily (community). Two nodes are similar if they are in the same densely connected region — even if structurally different (one is a hub, the other is a leaf).
DFS-like walks wander far into one community.

Structural equivalence (role). Two nodes are similar if they play the *same role* — both are hubs, both are bridges — even if in different communities.
BFS-like walks stay local and sample neighbourhoods.

Speaker notes

This distinction (homophily vs. structural equivalence) is one of the most important in network representation learning. It shows up across social network analysis, biology (proteins with similar function vs. same pathway), and recommendation (users with same taste vs. users at same stage of a purchase funnel). The two notions often pull embeddings in different directions; node2vec's p and q knobs let the practitioner commit to one or balance the two.

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Structural equivalence (role). Two nodes are similar if they play the *same role* — both are hubs, both are bridges — even if in different communities.
BFS-like walks stay local and sample neighbourhoods.

DeepWalk's uniform walks give a blur between the two. node2vec lets us choose.

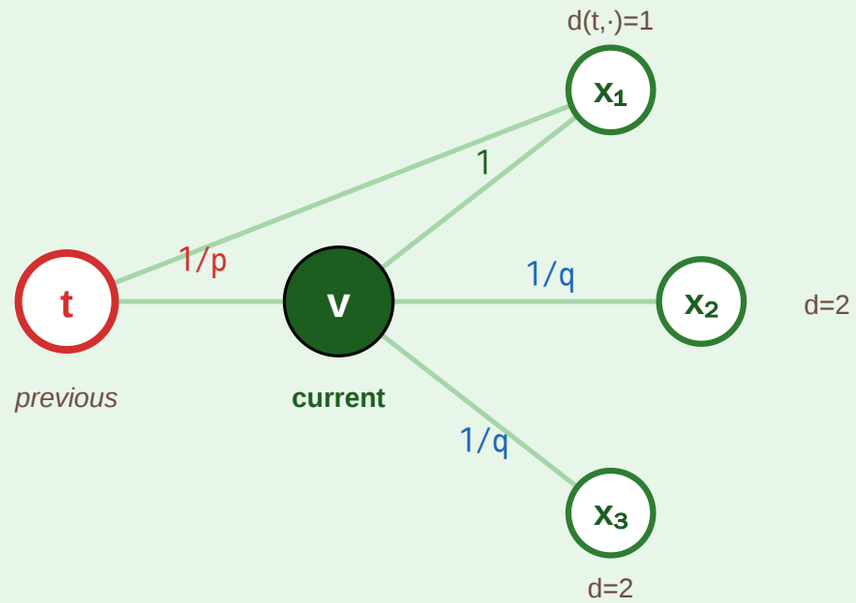
The Biased Walk

node2vec — Biased Random Walks with Return (p) and In-Out (q)

tune the walk between BFS-like (structural) and DFS-like (community) exploration

1. Second-order transition

biased by where we came from (t)



α(t → v → ·) based on distance d(t,·):

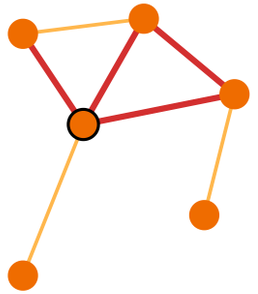
`d=0 (back to t): 1/p · d=1: 1 · d=2: 1/q`

2. Two regimes, two embeddings

p, q shape what "similar" means

small q → BFS-like

stay local, explore neighbours



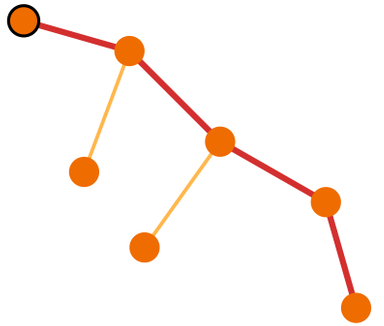
captures structural roles

hubs, bridges → similar z

"homophily by role"

small p → DFS-like

wander far, reveal communities



captures communities

same cluster → similar z

Grover & Leskovec 2016 — one knob, two inductive biases

Speaker notes

Given previous node t and current node v, the walk transitions to neighbour x with probability $\propto \alpha(t, x) \cdot w_{vx}$, where $\alpha = 1/p$ if $d(t, x) = 0$ (going back), $\alpha = 1$ if $d(t, x) = 1$, or $\alpha = 1/q$ if $d(t, x) = 2$. Small p keeps the walk local (BFS-like, structural); small q pushes it outward (DFS-like, community).

The bias is second-order — it depends on both the current and previous node, not just the current one. This is what lets the walk remember whether it is exploring locally or venturing further. Typical choices: $(p, q) = (1, 1)$ reduces to DeepWalk; $(p, q) = (4, 0.25)$ is strongly DFS-like; $(p, q) = (0.25, 4)$ is strongly BFS-like. In practice, grid-search or hyperparameter tuning finds good (p, q) for each dataset.

node2vec in Production

Despite its age, node2vec is still widely deployed because:

- **Scalable.** Runs on billions of edges with sampled walks + SGD.
- **No features required.** Works on raw adjacency, making it a sensible *baseline* for any graph task.
- **Interpretable knobs.** p and q correspond to clear exploration policies.

Speaker notes

node2vec is often the first method an industry team tries for a new graph-ML problem — it is robust, well-understood, and requires only the adjacency. But the moment node features are available (user profiles, product text, molecular fingerprints), GNNs usually dominate because they can fuse graph structure with features. Understanding the transition from random-walk to GNN is one of the main take-aways of this lecture.

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> **Limitation.** Like DeepWalk, node2vec is **transductive** and **feature-free**. A new node requires a new walk + retraining; node attributes (text, numeric features) are ignored. These shortcomings motivate GNNs.

Graph Neural Networks — Message Passing

Guiding Question: Can we learn node embeddings that *also* use node features — end-to-end with any downstream task?

Speaker notes

Graph Neural Networks (GNNs) are the modern answer to graph representation learning. They fix the three biggest limitations of spectral and random-walk methods: (1) they are *inductive* (new nodes can be embedded without retraining), (2) they use node features (not just adjacency), (3) they train end-to-end with the downstream task (node classification, link prediction, graph classification).

The Message-Passing Framework

At each layer l, every node v updates its representation by aggregating messages from its neighbours $\mathcal{N}(v)$:

$$h_v^{(l+1)} = \mathrm{UPDATE}\Big(h_v^{(l)}, \mathrm{AGG}\big(\{h_u^{(l)} : u \in \mathcal{N}(v)\}\big)\Big)$$

- AGG is a permutation-invariant function: sum, mean, max, or attention.
- UPDATE is usually a learnable linear transform + nonlinearity: $\sigma(W[h_v, m_v])$.
- Initial features $h_v^{(0)}$ can be node attributes, one-hot IDs, or pretrained embeddings.

Speaker notes

The message-passing formulation (Gilmer et al. 2017) is the unifying framework for essentially every GNN architecture published since. Different choices of AGG and UPDATE give different architectures (GCN, GraphSAGE, GAT, GIN, etc.), but the shape of the computation is universal. Understanding this framework once is enough to read 90% of the graph-ML literature.

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Stacking L layers gives each node a *receptive field* of L hops — its final embedding reflects its L-hop neighbourhood.

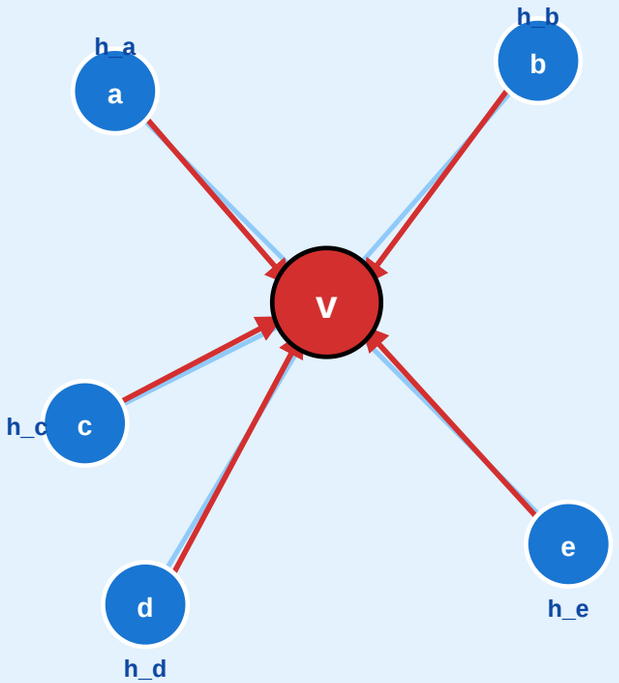
Message Passing — Visual

Graph Neural Networks — Message Passing

$h^{(l+1)}_v = \text{UPDATE}(h^{(l)}_v, \text{AGG}(\{ h^{(l)}_u : u \in N(v) \}))$

1. Neighbourhood of v

each node has a feature vector h_v

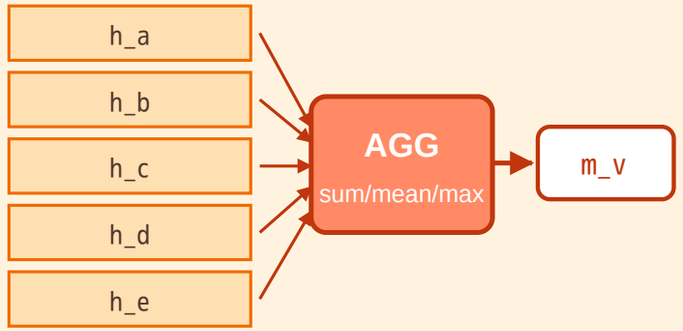


messages travel along edges

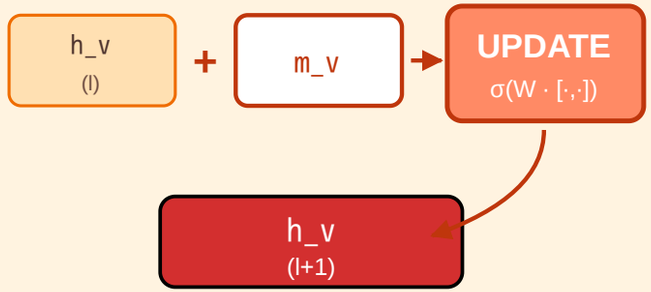
N(v) = { a, b, c, d, e }

2. Aggregate · Update

permutation-invariant pooling



then combine with own state



one GNN layer = one hop further

3. Stack L layers

receptive field = L hops

Layer 1 · 1-hop

Layer 2 · 2-hop

Layer 3 · 3-hop

Common variants

- GCN — normalised sum
- GraphSAGE — sampling
- GAT — attention weights
- GIN — sum + MLP

too many layers →
over-smoothing
all nodes collapse to same z

Speaker notes

(1) Each neighbour sends its current state as a message. (2) AGG pools the messages into a single vector m_v. (3) UPDATE combines m_v with the node's own previous state to produce $h_v^{(l+1)}$. Stacking layers extends the receptive field.

The figure shows the three conceptual steps of a single GNN layer. Read left-to-right: messages flow from neighbours, AGG pools them (sum/mean/max/attention), UPDATE combines with the previous state. The right panel shows that stacking L layers expands the receptive field — but more layers is not always better (see over-smoothing below).

Four GNN Variants in One Line Each

- **GCN** (Kipf & Welling 2017) — AGG = normalised sum: $\tilde{A} = D^{-1/2}(A + I)D^{-1/2}$; UPDATE = $\sigma(\tilde{A}HW)$.
- **GraphSAGE** (Hamilton et al. 2017) — AGG on a *sampled* subset of neighbours; scales to very large graphs; inductive by design.
- **GAT** (Veličković et al. 2018) — AGG weighted by learned *attention* α_{uv}.
- **GIN** (Xu et al. 2019) — AGG = sum; UPDATE = MLP. Provably as powerful as the Weisfeiler-Lehman graph isomorphism test.

Speaker notes

GCN is the canonical entry point — every graph-ML class starts there. GraphSAGE is the production workhorse (it scales to billion-node graphs because it samples neighbours rather than aggregating all of them). GAT adds attention, which tends to help when neighbour relevance varies. GIN is the theoretically most expressive variant. For most production problems, the *choice of features and training objective* matters more than the choice among these four.

The Over-Smoothing Problem

Stacking too many GNN layers kills the embedding.

- After L layers, each node's embedding averages over its L-hop neighbourhood.
- For random graphs, all nodes eventually see the entire graph $\rightarrow$ all embeddings collapse to the same vector.
- This happens already at $L = 3$–5 for typical benchmarks.

Speaker notes

Over-smoothing is the single most important pitfall of GNNs. Students see it as "more depth = more power" from CNNs and assume it transfers. It does not. A 2-layer GCN usually beats a 10-layer GCN on most node classification benchmarks. This is fundamentally different from computer vision — the graph topology limits how much depth helps. Understanding why is subtle: each layer mixes neighbour information with own information; after many layers, the "own information" is diluted.

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Remedies. Residual connections (JKNet), edge-weight normalisation, explicit node-pair features, or simply using fewer layers (1–3 is often enough).

Takeaway

GNNs learn node embeddings by repeated message passing — each layer extends the receptive field by one hop. They are inductive, feature-aware, and end-to-end trainable. But they suffer from over-smoothing: more layers is not always better.

Quick Check

You train a 2-layer GCN on a citation network to predict paper topics.

1. What is the receptive field of each paper's final embedding?
2. Why would stacking 10 layers hurt performance?
3. If you add a new paper tomorrow, can you predict its topic without retraining?

Speaker notes

Give them 3 minutes. (1) 2 hops — the embedding depends on the paper's direct citations and their citations. (2) Over-smoothing: after 10 layers each paper's embedding averages over essentially the entire graph, erasing local signal. (3) Yes — GCN (like GraphSAGE) is inductive. Feed the new paper's features and its known edges to the trained model for an embedding. This is the key advantage over DeepWalk/node2vec.

Closing the Loop — Embeddings Meet HNSW

Guiding Question: How do the embeddings we just built power modern retrieval systems — and what does this have to do with Kleinberg's grid?

Speaker notes

This module closes the L07–L09 arc. In L07 we met HNSW as a direct algorithmic descendant of Kleinberg's decentralized search — but we noted that HNSW needs *vectors* to index. This lecture has built exactly those vectors. The composition — embeddings + HNSW — is the backbone of every production retrieval system today.

The Full Pipeline

1. **Build embeddings** — spectral, DeepWalk, node2vec, or a trained GNN produces $z_v \in \mathbb{R}^d$ for each node.
2. **Build an index** — HNSW (L07) or similar ANN structure over $\{z_v : v \in V\}$.
3. **Query** — encode query q as z_q, return top-k nearest z_v in $O(\log n)$.
4. **Use** — link prediction, recommendation, RAG retrieval, semantic search.

Speaker notes

This is the "so what" slide. Students often see graph embeddings and ANN indexing as separate topics in two different classes. In production they are *always* used together: an embedding model produces dense vectors, an ANN index makes them searchable at scale. Understanding both — and understanding that the mathematical structure (Kleinberg's $r = d \leftrightarrow$ HNSW level distribution) is the same — is the single most valuable takeaway a network-science student can carry into industry.

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The same pipeline powers **product recommendation** (Alibaba, Amazon), **friend suggestion** (LinkedIn, Meta), **fraud detection** (PayPal), and **retrieval-augmented LLMs** (2024–2026 production stacks).

Why Geometry Matters

HNSW needs the embedding space to have the **right geometry** for greedy descent to work:

- **Similar nodes close, dissimilar nodes far** — true by construction in spectral / DeepWalk / GNN embeddings.
- **Low intrinsic dimension** — if embeddings live on a low-dimensional manifold inside $\mathbb{R}^d$, greedy search is efficient. Otherwise we hit the curse of dimensionality.
- **Smoothness** — continuous similarity (not sharp drops), which is why normalisation, temperature, and contrastive losses matter.

The *quality* of the embedding determines whether the index is actually navigable.

Speaker notes

This connects back to Kleinberg: his result said that the *exponent* r of the link distribution had to match the *dimension* d of the grid. For learned embeddings, we don't explicitly choose r — instead, the embedding method implicitly determines the distribution of distances between nodes, and HNSW adapts via its level-assignment geometric distribution. But the principle is the same: for greedy search to work in $O(\log n)$, the geometry has to be "navigable". A badly trained embedding (too uniform, or too collapsed) makes even the best ANN index useless.

Takeaway

Embeddings (L09) and navigable indexes (L07) are the two halves of the modern retrieval pipeline. Spectral, DeepWalk, node2vec, and GNN methods produce the vectors; HNSW descends them. Kleinberg's insight ($r = d$) reappears as an engineering guarantee in HNSW's layer structure.

Wrap-Up — The Representation Arc

Method	Uses	Scales to	Inductive	Feature-aware
Laplacian Eigenmaps	eigendecomposition	$\leq 10^6$ nodes	✗	✗
DeepWalk / node2vec	random walks + skip-gram	10^9 edges	✗	✗
GCN / GraphSAGE / GAT	message passing	10^9 edges	✓	✓
Combined w/ HNSW	any of the above → ANN index	10^9 vectors	✓	✓

Each row *adds* a capability without replacing the previous row — classical spectral embeddings remain the theoretical benchmark, random-walk methods remain the unsupervised baseline, GNNs add features, HNSW adds retrieval.

Looking Ahead: Exercise and Beyond

The exercise for this lecture compares Laplacian Eigenmaps, node2vec, and a small GCN on a shared dataset. Real graph-ML research in 2024–2026 has moved on to **graph transformers**, **graph foundation models**, and **heterogeneous GNNs** — but all rest on the message-passing core you now understand.

Reading

Chapter 23 of Hamilton's *Graph Representation Learning* (2020) covers spectral, random-walk, and GNN embeddings. Perozzi et al. (2014) is DeepWalk. Grover & Leskovec (2016) is node2vec. Kipf & Welling (2017) introduces GCN. Hamilton et al. (2017) introduces GraphSAGE. Malkov & Yashunin (2018) is the HNSW paper that closes the loop to L07.

Speaker notes

The comparison table is the conceptual spine of the lecture. Each row addresses a limitation of the previous row: Laplacian is principled but small; DeepWalk scales but has no features; GNNs add features; HNSW adds retrieval. In the exam, students should be able to (a) explain what each method captures, (b) pick the right method for a given scenario, and (c) explain why the full pipeline (embedding + HNSW) is the backbone of modern graph applications.